S181M116 Derivatives and Alternative Investments
Final Exam, Variant B: Multiple Choice

Time allowed: 1 hour 30 minutes   Total: 80 points

Instructions. Answer all questions. Each item has exactly one best answer. Unless stated otherwise, ignore transaction costs, taxes, bid-ask spreads, and default risk. Use continuous compounding. Round monetary answers to two decimals, probabilities to four decimals, and hedge ratios to four decimals.

Scoring. There are 20 multiple-choice items. Each item is worth 4 points, for a total of 80 points.

Coverage. This exam is based on the material covered in class: derivative basics, market structure, forwards and futures, hedging and basis risk, forward/futures pricing, options, put-call parity, binomial trees, and option no-arbitrage restrictions. Lecture 3 and Lecture 5 material are excluded.

Question 1: Concepts, Payoffs, and Market Structure [16 points]

This question is on the easier side.

1. A contract pays $\max(90 - S_T, 0)$ in six months. Which statement is correct? [**4 points**]
   - A. It is not a derivative because it has no initial premium.
   - B. It is a derivative whose underlying variable is the asset price S_T.
   - C. It is a bond because the payoff is bounded above.
   - D. It is a futures contract because the payoff is linear.

2. A European call and a European put both have strike $K = 75$. The call premium is 6 and the put premium is 4. If $S_T = 88$, what are the buyer's call payoff, call profit, put payoff, and put profit? [**4 points**]
   - A. $13,\ 7,\ 0,\ -4$
   - B. $13,\ 13,\ 0,\ 0$
   - C. $0,\ -6,\ 13,\ 9$
   - D. $7,\ 13,\ -4,\ 0$

3. Which statement best describes an OTC derivative relative to an exchange-traded derivative? [**4 points**]
   - A. OTC contracts are more standardized and always centrally cleared.
   - B. OTC contracts are customizable, but can create more counterparty, valuation, and liquidity risk.

C. OTC contracts have no counterparty risk because they are private.
D. OTC contracts are always cheaper because they have no margin.

4. Why can margin create liquidity pressure? [**4 points**]
   A. Because daily settlement can require losing positions to post cash before the underlying exposure has generated offsetting cash flow.
   B. Because margin is the purchase price of the futures contract.
   C. Because variation margin removes all price risk.
   D. Because margin calls occur only when the hedge is profitable.

Question 2: Futures Settlement and Forward Pricing [16 points]

This question is on the easier side.

1. A trader is **short** eight copper futures contracts. Each contract is for 25,000 pounds. Yesterday's settlement price was USD 4.12 per pound and today's settlement price is USD 4.05 per pound. What is today's futures profit or loss? [**4 points**]
   A. USD 14,000 gain
   B. USD 14,000 loss
   C. USD 1,400 gain
   D. USD 1,400 loss

2. The trader's margin account balance before settlement was USD 38,000. The maintenance margin is USD 30,000 and the initial margin is USD 38,000. What is the margin account balance after settlement, and is there a margin call? [**4 points**]
   A. USD 24,000; yes, deposit USD 14,000.
   B. USD 30,000; no margin call.
   C. USD 52,000; no margin call.
   D. USD 52,000; yes, deposit USD 8,000.

3. A stock has current price $S_0 = 120$. The present value of known cash dividends during a six-month forward contract is $I = 2.40$. The continuously compounded risk-free rate is 3.5% per year. What is the no-arbitrage forward price? [**4 points**]
   A. 117.60
   B. 119.68
   C. 122.10
   D. 124.25

4. A dealer quotes a forward price of 121 for the same contract. Which statement is correct? [**4 points**]
   A. The quote is too low; short the asset and long the forward.
   B. The quote is too high; buy the asset, short the forward, and lock in about 1.32 per share.
   C. The quote is exactly no-arbitrage.
   D. The quote is too high; short the asset, invest proceeds, and long the forward.

Question 3: Cross Hedging, Basis Risk, and Effective Prices [16 points]

This question is on the harder side.

A manufacturer will buy 1,250,000 pounds of copper in three months. It cross hedges with a related metals futures contract. Each futures contract covers 25,000 pounds. Historical data give

$$\rho = 0.72, \qquad \sigma_S = 0.18, \qquad \sigma_F = 0.16.$$

The hedge is opened at $F_1 = 3.80$. At hedge close, the manufacturer buys copper at $S_2 = 4.12$ and closes futures at $F_2 = 4.05$.

1. What hedge direction should the manufacturer use? [**4 points**]
   - A. Long futures, because it will buy the commodity later and wants protection against rising prices.
   - B. Short futures, because it will buy the commodity later.
   - C. Long futures only if the expected basis is negative.
   - D. No hedge is possible because the contract is a cross hedge.
2. What is the minimum-variance hedge ratio? [**4 points**]
   - A. 0.6400
   - B. 0.7200
   - C. 0.8100
   - D. 1.1250
3. What number of futures contracts is implied by the hedge ratio before rounding? [**4 points**]
   - A. 36.0
   - B. 40.5
   - C. 50.0
   - D. 56.25
4. What is the effective purchase price per pound, and how does the realized basis compare with an expected basis of 0.03? [**4 points**]
   - A. 3.87; the realized basis is worse for the buyer.
   - B. 3.87; the realized basis is better for the buyer.
   - C. 4.37; the realized basis is worse for the buyer.
   - D. 4.37; the realized basis is better for the buyer.

Question 4: Option No-Arbitrage and Binomial Pricing [16 points]

This question is on the harder side.

Consider European options on a non-dividend-paying stock. The current stock price is $S_0 = 48$, the strike is $K = 50$, the continuously compounded risk-free rate is $r = 5\%$, and maturity is $T = 0.25$ years. For the parity questions, a European call trades at $c = 2.60$. Separately, for the binomial model questions, use $u = 1.18$ and $d = 0.88$.

1. What is the no-arbitrage European put price implied by put-call parity? [**4 points**]
   - A. 1.22
   - B. 2.60
   - C. 3.20
   - D. 3.98
2. Suppose the market put price is $p = 3.20$. Which arbitrage statement is correct? [**4 points**]
   - A. The protective put is expensive; sell it and buy the fiduciary call.
   - B. The fiduciary call is expensive; sell it and buy the protective put.
   - C. Both packages are fairly priced.
   - D. There is no arbitrage because call and put prices are both positive.
3. In the one-step tree, what are the up-state and down-state stock prices and call payoffs? [**4 points**]
   - A. $S_u = 56.64$, $S_d = 42.24$; payoffs 6.64 and 0.
   - B. $S_u = 56.64$, $S_d = 42.24$; payoffs 0 and 7.76.
   - C. $S_u = 59.00$, $S_d = 44.00$; payoffs 9.00 and 0.
   - D. $S_u = 48.18$, $S_d = 47.88$; payoffs 0 and 0.
4. Which pair is closest to the risk-neutral probability and binomial call value? [**4 points**]
   - A. $q = 0.4419$, call value 2.90
   - B. $q = 0.4419$, call value 6.64
   - C. $q = 0.5581$, call value 2.60
   - D. $q = 0.5581$, call value 3.67

Question 5: Option Bounds, Early Exercise, and Strategy Logic [16 points]

This question is on the harder side.

Consider options on a non-dividend-paying stock with current price $S_0 = 70$, strike $K = 68$, risk-free rate $r = 4.5\%$ with continuous compounding, and maturity $T = 0.75$ years.

1. What is the lower bound for the European call, and what does it imply if the call is quoted at $c = 3.80$? [**4 points**]
   - A. Lower bound about 0; no arbitrage.
   - B. Lower bound about 1.74; no arbitrage.

C. Lower bound about 4.26; the call violates the lower bound.

D. Lower bound about 6.00; the call is fairly priced.

2. What is the lower bound for the European put, and what does it imply if the put is quoted at $p = 0.60$? **[4 points]**

   A. Lower bound 0; no lower-bound arbitrage from this quote.

   B. Lower bound 1.74; the put violates the lower bound.

   C. Lower bound 4.26; the put violates the lower bound.

   D. Lower bound 68; the put violates the lower bound.

3. If a European call with this strike and maturity trades at $c = 6.00$, what is the no-arbitrage European put price from put-call parity? **[4 points]**

   A. 0.00

   B. 1.74

   C. 3.80

   D. 6.00

4. Which statement is correct? **[4 points]**

   A. An American put price of $P = 1.50$ is consistent if the European put value from parity is about 1.74.

   B. A covered call is risk-free because the writer owns the stock.

   C. An American put must be worth at least as much as the otherwise identical European put, and a covered call still has downside stock risk.

   D. A covered call has unlimited upside because the writer owns the stock.